

SPB traction inverter specification

$P = 150 \text{ kW}$, $V_{\text{bus}} = 1200 \text{ V}$, $M \leq 1.15$, $\eta \geq 99.5 \%$, $T_{j,\text{max}}$
 $\Rightarrow I_{\text{ph}} = 100 A_{\text{rms}}$ for any number of submodules N

Machine side (multiphase, N winding sets)

ripple $\Delta i \propto V_m / (L_\sigma f_{\text{sw}})$, winding step V_m , dv/dt , $R_{\text{ac}}(f)$
 $\Rightarrow N$, V_m , f_{sw} ; integration: space, coolant, end winding

$10V_m$

f_{sw}

$f_{\text{sw, ripple}}$

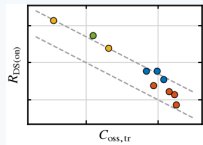
Voltage class & device

GaN datasheets
 $R_{\text{DS(on)}}, Q_{\text{oss}}, V_{\text{DS}}$

classes 100, 200, 350, 650 V
 V_m , $N = V_{\text{bus}}/V_m$

$P_{\text{min}} = 6IV_{\text{bus}}\sqrt{f_{\text{sw}}\tau}$
 $\tau = R_{\text{DS(on)}}C_{\text{oss,tr}}$

EPC2361, $V_m = 75 \text{ V}$, $N = 16$



Section III

EPC2361 model, V_m

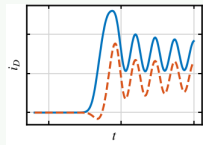
Layout & gate drive

sharing sensitivity, 2 FETs
 $L_G, L_{CS}, L_D, V_{\text{th}}$ (LTspice)

equal L_G and L_{CS} :
 star driver per FET pair, Kelvin

two mirrored cells,
 vertical loops on L1/L2

cell + driver concept, scaling



Section IV

geometry, driver

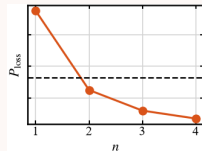
Devices in parallel

$n = 1, 2, 3, 4$ per switch

$P_{\text{cond}} \propto 1/n$, $P_{\text{sw}}(n)$, P_{Cu}

loss gain vs.
 drivers $n_{\text{drv}}?$

$n = 2$ (one driver)



Section V

n

DC-link capacitors

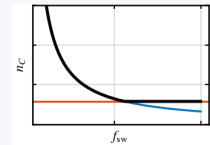
$\Delta v_{\text{pp}} \rightarrow C_v$

$I_{C,\text{rms}} \rightarrow C_i$

$C_{\text{dc}} = \max\{C_v, C_i\}$

MLCC: $C(V_{\text{dc}})$, ESR, ESL,
 local + bank hierarchy

$24 \times 1210 + 12 \times 0805$



Section VI

$C, \text{ESR}, \text{ESL}$

Verification with one parametric geometry

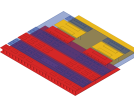
Sections VII–XI

PCB placement



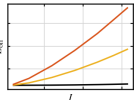
KiCad, 3D models

Q3D extraction



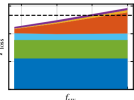
$R(f), L(f)$,
 $L_{\text{loop}}, L_G, L_{CS}$

LTspice



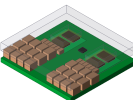
$E_{\text{on}}, E_{\text{off}}, R_G$,
 $V_{\text{DS}}, \text{gate bump}$

Loss model



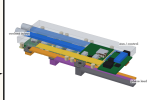
cond., sw., $\text{Cu}(f)$, MLCC

Icepak



T_j, T_{PCB} , cooling

Cell assembly



cold plate, busbar,
 insulation, leads

$\hat{V}_{\text{DS}}, \eta, T_j$, sharing,
 machine loss met?

yes

prototype: double pulse,
 continuous, motor (future work)

no: modify layout, $R_G, n, C_{\text{dc}}, f_{\text{sw}}$